

CSD 598 - Winter 2026

Data exploration:
Descriptive statistics and data visualisation

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3 Feb 2026

Last time: Ethics

Any questions? Any thoughts?

This time: Data exploration

- ▶ What makes statistics?
- ▶ Why do data exploration?
- ▶ Descriptive stats
- ▶ Connecting descriptive stats and visualisation via dependence of variables
- ▶ Some principles of and resources for visualisation
- ▶ An example – if we have time

What makes statistics?

- ▶ Logic
- ▶ Maths
- ▶ Computation
- ▶ Relative to the question and field

And here a gentle reminder: There is no such thing as unpolitical data.

Why do data exploration?

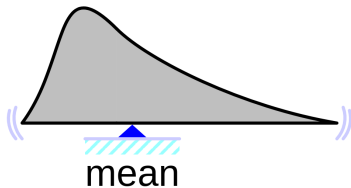
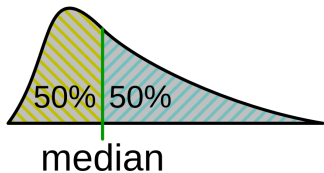
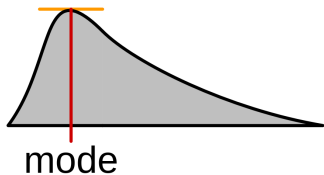


"Remember kids, the only difference between screwing around and science is writing it down."

Adam Savage, quoting ballistics expert Alex Jason

Descriptive stats

Central tendencies



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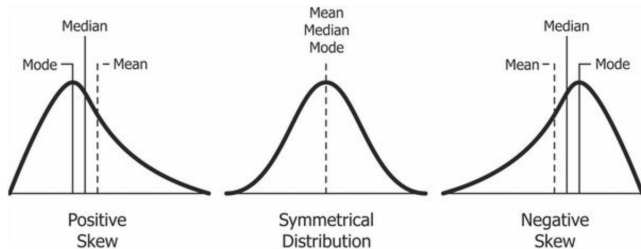
Dispersion and shape

Dispersion

- ▶ Range, quartiles: easy to calculate on samples
- ▶ Standard deviation (SD), variance: most used
- ▶ Average absolute deviation: more robust than SD

Shape measures

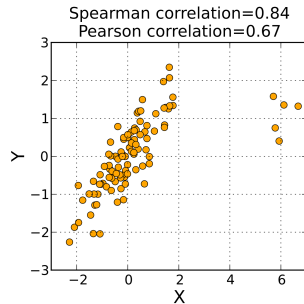
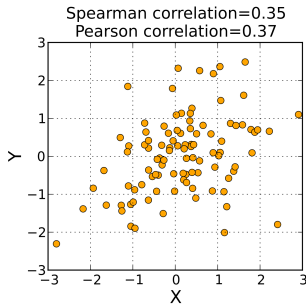
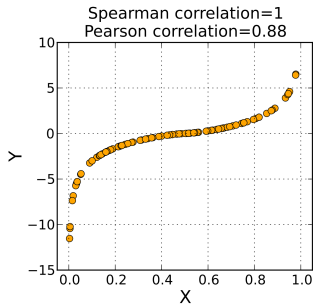
- ▶ Skewness: which way does the distribution lean?
- ▶ Kurtosis: tailedness, things get complex here



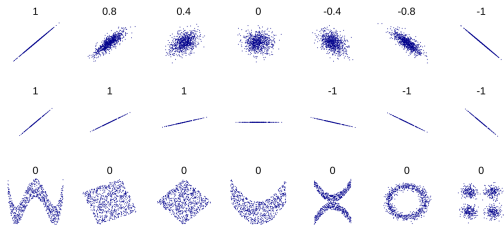
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Dependence between variables: The usual

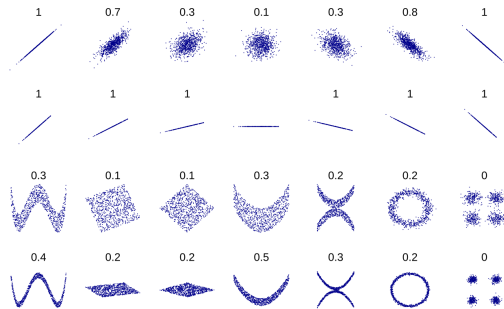
- Covariance: measure of the linear relationship between two variables.
- Pearson's ρ : correlation of continuous variables
- Spearman's ρ : correlation of ordering of variables



Dependence between variables: Pearson vs Distance correlation



Pearson correlation



Distance correlation

Denis Boigelot, CC BY-SA 3.0, via Wikimedia Commons

Visualisation



Dr. Lindsay Portnoy
@lportnoy

This is not how pie charts work



9:48 PM · Mar 26, 2020 · [Twitter for iPhone](#)

8.1K Retweets 49.8K Likes

Choosing a good graph

- ▶ The following need to serve a purpose:
 - ▶ Graph type
 - ▶ Colour
 - ▶ Fancy effects
- ▶ Generally, the purpose should be conveying information.
- ▶ Secondly, it can be to aid with conveying of information. For example, by making the graph look nice and actually be read as a result.

Designing a good graph

- ▶ Figure out what you are wanting to say and how the data shows that.
- ▶ Printed versions may be very different from what is shown on a screen.
- ▶ Mind accessibility:
 - ▶ Everybody does not see colour in the same way.
 - ▶ Everybody does not see.

Resources for choosing a graph

- ▶ <https://royal-statistical-society.github.io/datavisguide/>
Good, approachable guide to data visualisation.
- ▶ <https://www.data-to-viz.com/>
Actual recommendations with actual code (R, Python, D3.js, React) to get going.
- ▶ <https://r-graph-gallery.com/>
Neat gallery for getting to know different graph types. (Not exhaustive though.)
- ▶ Doesn't take much to find more. Please post any good ones you like in Canvas.
- ▶ My own favourite tool is the pairs plot in R. So we'll have a look at an example of that.

Exploration example

- ▶ This comes from the latest data analysis I have run.
- ▶ As it is unpublished, I ask it not be recorded, and hence will not describe it here on the slides.
- ▶ I am happy to talk about it in detail and show how I've done the whole thing.